BIOS 755: Missing Data Wrap-up

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Missing Data: Overview

Two approaches:

- Likelihood-based (Covariance Pattern Analysis/ LMM / GLMM)
- GEE-based (standard / weighted)

Key idea: Validity depends on missingness mechanism

Likelihood Methods: MCAR / Covariate

- **MCAR:**
  - Valid
  - Use standard model
- **Covariate-dependent missingness:**
  - Valid if covariates (appropriately) included

Likelihood Methods: Baseline / MAR

- **Missing baseline covariates:**
  - Must be handled separately (e.g., imputation)
- **MAR:**
  - Valid if model correctly specified
  - Uses all available data
  - Imputing missing outcomes is more robust to model misspecification

Likelihood Methods: MNAR

- **MNAR:**
  - Not valid
  - Requires sensitivity analysis or alternative models

GEE Methods: MCAR / Covariate

- **MCAR:**
  - Standard GEE is valid
- **Covariate-dependent missingness:**
  - Valid if covariates included

GEE Methods: Baseline / MAR

- **Missing baseline covariates:**
  - Must be handled separately (e.g., weighting)
- **MAR:**
  - Standard GEE not valid
  - Use weighted GEE (IPW)

GEE Methods: MNAR

- **MNAR:**
  - Not valid
  - Requires sensitivity analysis or specialized methods

Key Takeaways

- Likelihood: valid under MAR (if correctly specified)
- GEE: needs weights under MAR
- Baseline covariates missing $\rightarrow$ handle separately
- MNAR $\rightarrow$ no standard solution